

INTERCEPT

Names: _____

Period: 0

Date: _____

UNIT 2: PIECEWISE 1D MOTION, REST INTERVALS & PURSUIT SLOPES

SURVEILLANCE DOSSIER: DESERT HIGHWAY RECONNAISSANCE

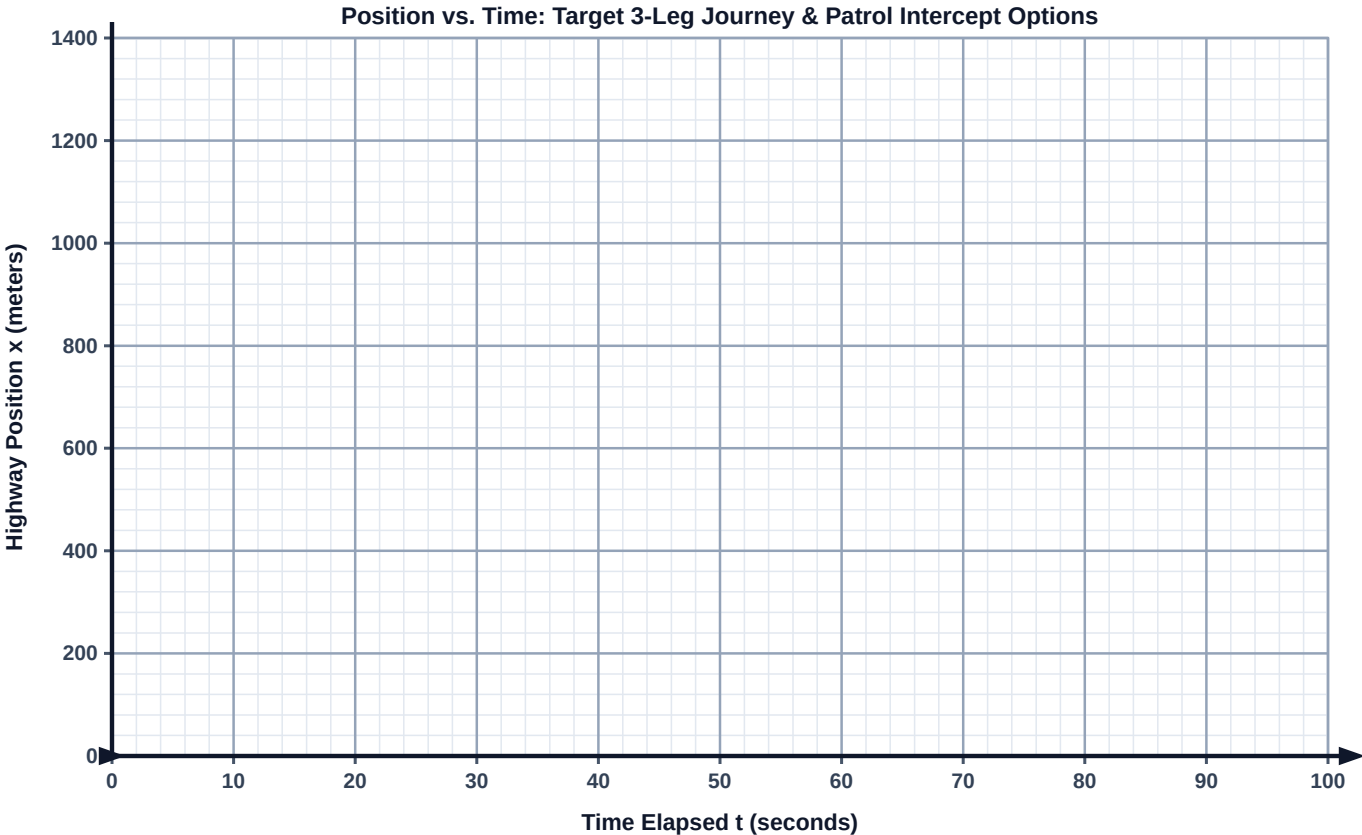
A target vehicle heads east on a straight highway starting from $x = 0\text{ m}$ at $t = 0\text{ s}$ in 3 distinct legs:

- **Leg 1 ($t = 0$ to 20 s):** Drives at **15.0 m/s** (covers $15\text{ m/s} \times 20\text{ s} = 300\text{ m}$, reaching $x = 300\text{ m}$ at $t = 20\text{ s}$).
- **Leg 2 ($t = 20$ to 50 s):** Pulls over at a rest stop and remains **completely stopped ($v = 0\text{ m/s}$)** for 30 seconds.
- **Leg 3 ($t = 50$ to 90 s):** Re-enters the highway at **25.0 m/s** for 40 s (covers $25\text{ m/s} \times 40\text{ s} = 1,000\text{ m}$, reaching $x = 1,300\text{ m}$ at $t = 90\text{ s}$).

A Highway Patrol Cruiser is at $x = 0\text{ m}$, but is held at the station and **cannot leave until $t = 30\text{ seconds}$** .

 **Graphing Task:** 1) Plot Target's 3-leg line (0 to 90s). 2) Draw Patrol Line A (from $t=30\text{s}$, $x=0$ to catch at rest stop at $t=50\text{s}$). 3) Draw Patrol Line B (to catch at $t=90\text{s}$).

SLOPE AS
VELOCITY



GRAPHICAL ANALYSIS & PURSUIT STRATEGY

ANSWER USING YOUR GRAPH ABOVE

1. Interpreting the Rest Stop (Leg 2):
What does the visual slope of the line represent between $t = 20\text{ s}$ and $t = 50\text{ s}$? How does the graph show the car is stopped?

2. Cruiser Speed Calculations (From Slopes of Lines A and B):

- **Strategy A (Catch at rest stop at $t = 50\text{ s}$):** Cruiser travels 300 m in 20 s (from $t=30$ to 50s). Required speed: _____ m/s
- **Strategy B (Catch at $t = 90\text{ s}$):** Cruiser travels 1,300 m in 60 s (from $t=30$ to 90s). Required speed: _____ m/s

3. Speed Limit Constraint Analysis:
The patrol cruiser has a top safe speed cap of **20.0 m/s (45 mph)**. Which of the two strategies is physically possible? Defend with calculations: